

SMALLER PV PLANTS MUCH BETTER

Giving importance to solar-generated power has lot of benefits including improvement in power quality, minimising voltage swings within limits and elimination of blackout for consumers at the terminal ends



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The government of India is implementing solar energy utilisation in a very ambitious manner. The target for 2012 has been set at over 2 gigawatt. It appears difficult to achieve this target in spite of the efforts being made. The actual installed grid-fed solar power generation, both from solar thermal power as well as photovoltaic (PV) solar systems is much below the expected level.

However, the government needs to be congratulated for taking this bold decision and setting up the solar mission. Even two-three gigawatt of solar energy can be gainfully used to give cleaner power to electrical consumers. PV solar power, even though a small contributor to our huge energy shortfall, has features that will allow us to give clean power to consumers and farmers during the day time. This

can be done by allowing small and medium enterprises (SMEs) to get into solar power generation business by installing 50-300 kilowatt-peak units (kWp) solar power plants primarily using unused roof-tops and terraces in local vicinity.

If such plants feed their output into a local substation, it will bring about instant improvement in the power quality in various towns and rural farming areas during the daytime when the power shortage is most severe. This will help the country to mitigate the power problem faced in many areas and replace diesel generators and home inverters.

Current electrical power supply status

International Energy Agency data for 2008 indicates that electrification rate for India was nearly 65 per cent for the country as a whole. In urban areas 93 per cent had access to electricity com-

pared to rural areas, where electrification rates were approximately 50 per cent. Roughly 400 million people did not have access to electricity in India. The country had approximately 170 gigawatt of installed electric capacity and generates 830 billion kilowatt-hour (kWh) of energy, nearly all of it from coal, oil or gas.

Conventional thermal sources produced over 80 per cent of electricity. Hydro-electricity accounted for nearly 16 per cent and nuclear energy roughly 2 per cent of electricity. Balance 2 per cent was from renewable sources like wind and solar.

Today, total electrical energy available from all sources is 830 gigawatt-hour (GWh). The demand is much larger and, as a result, most of the Indian consumers, who are lucky to receive any power at all, get it of very poor quality. There are frequent blackouts often lasting several hours at a time. When power is available it suffers from deep voltage swings; from as low as 140V to as high as 270V. Besides the shortage in generation, the grid itself is poor in performance, resulting in huge power losses estimated to be over 25 per cent due to line-drops and thefts. Indian consumers are essentially the victims of poor power planning by successive governments at the centre.

PV solar energy utilisation merits

There are several benefits of using solar power:

1. Free energy input is available in abundance. In one hour the sun delivers more energy to earth than the global consumption in a year.
2. Solar is the safest renewable energy resource and available during power shortage time.
3. Current PV solar energy cost per kWh is less than Rs 8.50 and reducing.
4. Small-capacity solar power plants of 20- to 100-kilowatt capacity are as cost-effective as large plants of 5- to 20-megawatt capacity and thus ideal for distributed generation.
5. Time to commission 20-200 kilo-